UNIVERSITY OF CENTRAL PUNJAB

(Faculty of IT)

Computer Communications and Networks (CCN)

Assignment No 3 **Date : 14 May 2026**

Total Marks: 100 **Due Date : 20 May 2026**

CLO-3 Questions

1. Consider the diagrams of TCP and UDP header structures with their various fields (draw the diagrams in your answer). Compare the functionality of TCP and UDP protocols, based on the differences between their header fields; i.e, what additional header fields are in TCP compared to UDP, and the added functionality they provide. (comparison is to be brief).

2. Given are two measured sample RTT values, as 110 ms and 180 ms. Compare the values of the TCP Timeout Intervals, computed after each sample. Assume the value of estimated RTT as 103 ms, and value of Dev RTT as 6 ms, just before the first of these samples were obtained. Take the value of $\alpha = 0.125$ and $\beta = 0.25$ for these calculations.

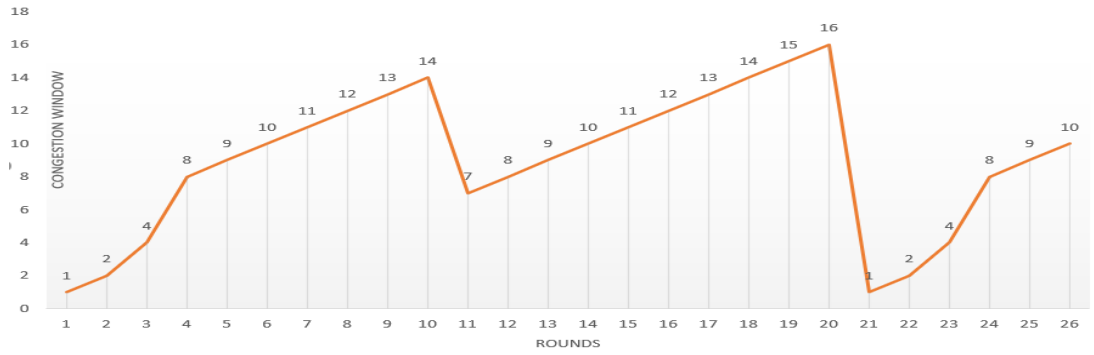
3. Examine the actions taken in each of following scenarios in TCP, at sender or receiver side:-

   (a) Segments with sequence numbers 1, 2, 3, 4 & 5 are sent back to back. If sender is receiving ACK 2 again and again, what is the meaning of that. What action will sender perform.

   (b) Host A sends 3 TCP segments to Host B, after connection has been established; asssume ISN is 200. How do the sender and receiver assign sequence numbers and acknowledgement numbers to these segments containing the following data (show in a timing diagram).

   Segment 1: 20 bytes Segment 2: 40 bytes Segment 3: 15 bytes

4. Examine the behavior of TCP Reno in the following figure, and answer the given questions. In all cases, you should provide a short discussion justifying your answer :-



   (a) Identify intervals of time when TCP slow start is operating; why SS is needed here.

   (b) Identify intervals of time when TCP congestion avoidance is in operation; why CA needed here.

   (c) Identify the time when fast recovery is applied; why do we need fast recovery here.

   (d) Identify the time where packet loss is due to timeout.

   (e) How many ssthreshold values are used here; write all ssthreshold values and their intervals.

   (f) Is there any triple duplicate ACK event; if yes, identify the time interval.

5. Analyze the effect of using Slow Start on a link with a 10 msec RTT. The receiver window size (rwnd) is 17 KB and MSS is 1 KB. Data transfer starts with SS phase, during which a packet drop event occurs (triple duplicate ACK) when congestion window (cwnd) became 16 KB. After that there is no other packet drop event.

   (a) Draw the graph for this connection, showing change of cwnd size with RTTs.

   (b) For how much time the connection was in SS phase; answer in terms of RTTs, and also in milli-seconds.

   (c) How long does it take after the packet drop event till the time sender is able to send full rwnd amount of data; answer in terms of RTTs, and also in milli-seconds. What is the average thruput achieved during this time.

Assignment Guidelines:

Assignment is to be done ***individually, handwritten,*** *and submitted in* ***hard copy****.*

Assignment should have a ***separate title page****; mention your Name, ID, Section and Assignment number.*

Assignments will ***NOT be accepted after the due date.***

Marking Criteria: *50% marks will be awarded for complete submission of assignment; 50% evaluation will be based on Assignment Test, to be conducted in class.*